

DYNAMICS & CONTROL 3

CONTROL

Section:

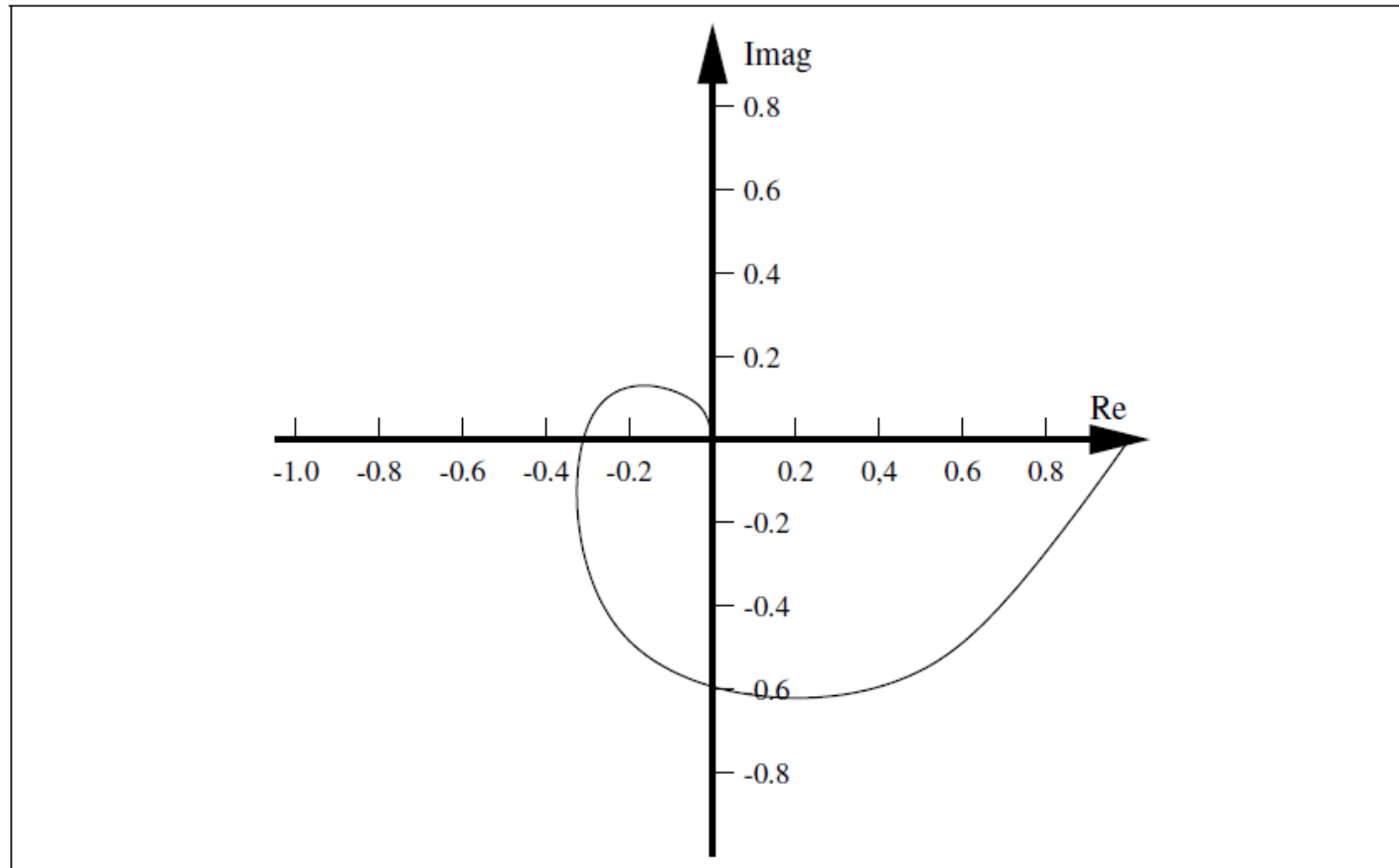
Exercises

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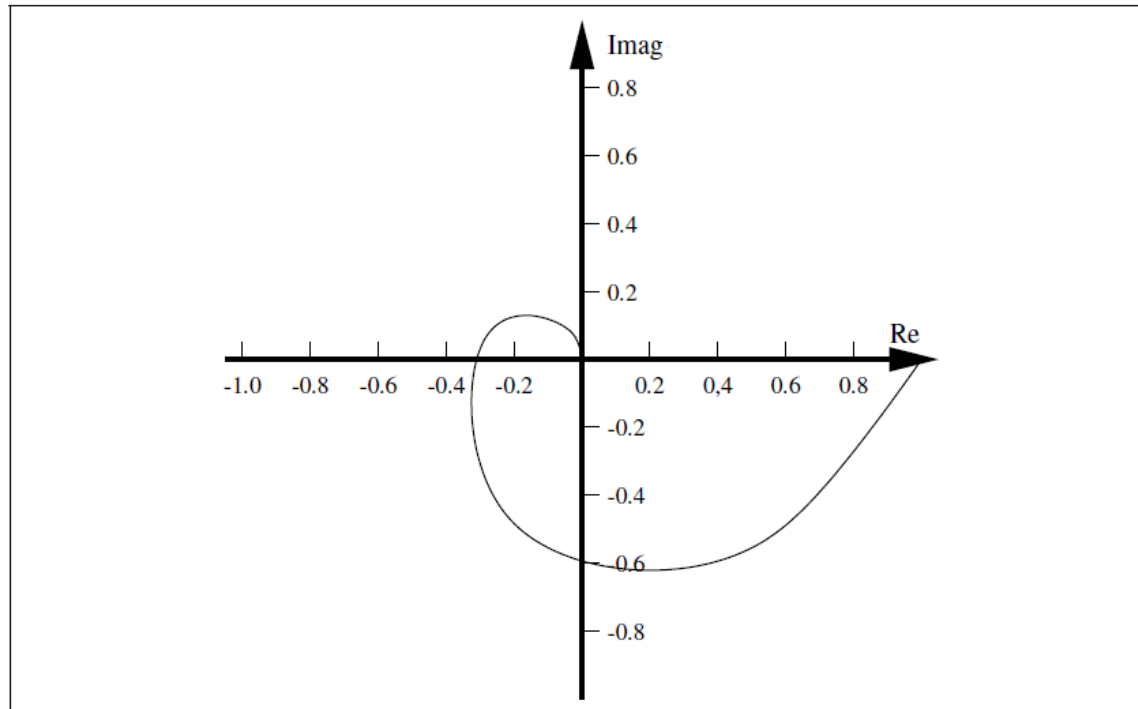
Worked example 1

1. Experimental data from a stable open-loop system is shown in the Nyquist plot.



Prove that the closed-loop system with unity negative feedback is also stable. Comment on the stability if the magnitude is increased by 20dB (a factor of 10)

Worked example 1



Solutions: The locus does not encircle the critical point, -1, as it crosses the negative real axis at -0.3. Therefore, the closed-loop system is stable.

If the magnitude is increased by 20dB (which is 10 in linear), the crossing point changes from -0.3 to -3. This time the locus will encircle the critical point -1 and the closed-loop system is now unstable.

Worked example 2

A control system has an open-loop transfer function

$$G(s) = \frac{5}{s \left(1 + \frac{s}{10}\right)}$$

- (a) Find the limit of $Re[G(j\omega)]$ as $\omega \rightarrow 0$? Sketch the Nyquist diagram and find the phase margin.
- (b) Why is there no point asking you to calculate the gain margin?
- (c) Sketch the Bode diagram for this system.

Solutions:

(a) The frequency response of the open loop is:

$$H(s) = \frac{5}{s \left(1 + \frac{s}{10}\right)} \Rightarrow H(j\omega) = \frac{5}{j\omega \left(1 + \frac{j\omega}{10}\right)} = \frac{5}{-\frac{\omega^2}{10} + j\omega}$$

Let's extract the real and imaginary parts:

$$\operatorname{Re}[H(j\omega)] = -\frac{\frac{1}{2}}{1 + \frac{\omega^2}{100}}$$

$$\operatorname{Im}[H(j\omega)] = -\frac{5\omega}{\frac{\omega^4}{100} + \omega^2}$$

We can also get the gain of H:

$$|H(j\omega)| = \frac{5}{\sqrt{\frac{\omega^4}{100} + \omega^2}}$$

For the phase, we should note first that the locus plot is in the third quadrant (both the real and imaginary parts are negative for all positive frequencies), therefore the phase is:

$$\angle H(j\omega) = \tan^{-1}\left(\frac{10}{\omega}\right) - 180^\circ$$

To sketch the locus plot, we can look at the following points: 0, 10 (corner frequency of the lag term) and $+\infty$.

- As 0 is a pole of H , let's look at the limit of both real and imaginary parts when $\omega \rightarrow 0$:

$$\operatorname{Re}[H(j\omega)] \rightarrow -\frac{1}{2} \text{ when } \omega \rightarrow 0$$

$$\operatorname{Im}[H(j\omega)] \rightarrow -\infty \text{ when } \omega \rightarrow 0$$

So, when $\omega \rightarrow 0$, the Nyquist plot has a vertical asymptote at $\operatorname{Re} = -0.5$.

- $\omega \rightarrow +\infty$: as we would expect the gain to tends toward 0, let's look at the phase (and verify that our gain assumption is correct).

$$|H(j\omega)| \approx \frac{50}{\omega^2} \rightarrow 0$$

$$\angle H(j\omega) \rightarrow -180^\circ$$

- $\omega = 10$:

$$|H(j10)| = \frac{5}{\sqrt{200}} = \frac{1}{2\sqrt{2}}$$

$$\angle H(j10) = -135^\circ$$

$$\operatorname{Re}[H(j10)] = -\frac{1}{4}$$

$$\operatorname{Im}[H(j10)] = -\frac{1}{4}$$

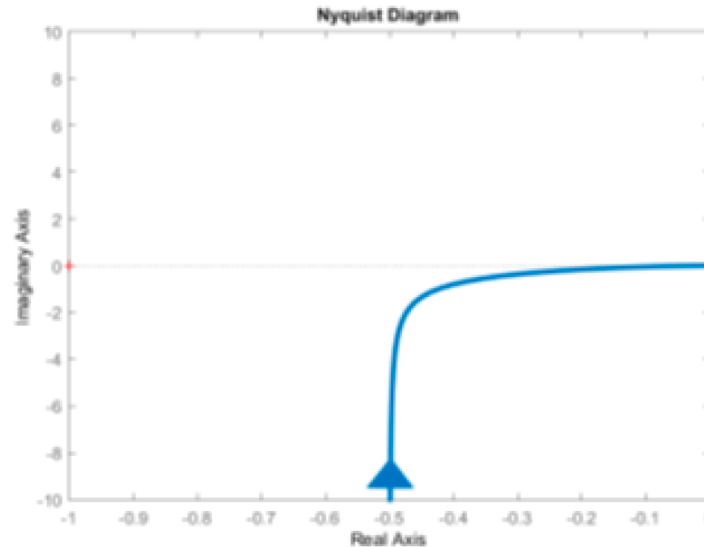


Figure 2a: Nyquist plot for H.

The phase margin is calculated from the phase at the frequency such that $|G(j\omega)| = 1$. The definition is:

Let's note ω_p the frequency that gives $|G(j\omega_1)| = 1$ and θ the phase at this frequency, i.e. $\theta = \angle G(j\omega_1)$. The phase margin, Φ , is defined by:

$$\Phi = 180 - \theta$$

To find the phase margin, we can follow this process:

- 1- Find frequencies that gives $|G(j\omega)| = 1$ and select the positive frequency;
- 2- Calculate the phase of $G(j\omega)$ for the frequency found in 1;
- 3- Deduce the phase margin of the system.

The gain of the open loop transfer function is:

$$|H(j\omega)| = \frac{5}{\sqrt{\frac{\omega^4}{100} + \omega^2}}$$

We are looking for ω that gives:

$$|H(j\omega)| = \frac{5}{\sqrt{\frac{\omega^4}{100} + \omega^2}} = 1$$

We can organise this equation by taking the square and do a cross-multiplication:

$$\frac{\omega^4}{100} + \omega^2 - 25 = 0$$

This equation has one positive real solution: $\omega_1 = 4.5509$.

Let's calculate the phase at this frequency. We have:

$$\theta = \angle H(j\omega_1) = \tan^{-1}\left(\frac{10}{\omega_1}\right) - 180^\circ = -114.47^\circ$$

The phase margin is:

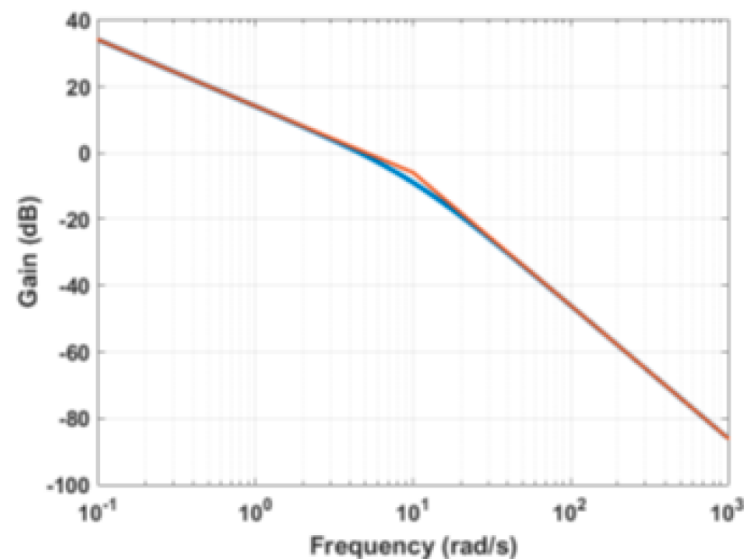
$$\Phi = +65.53^\circ$$

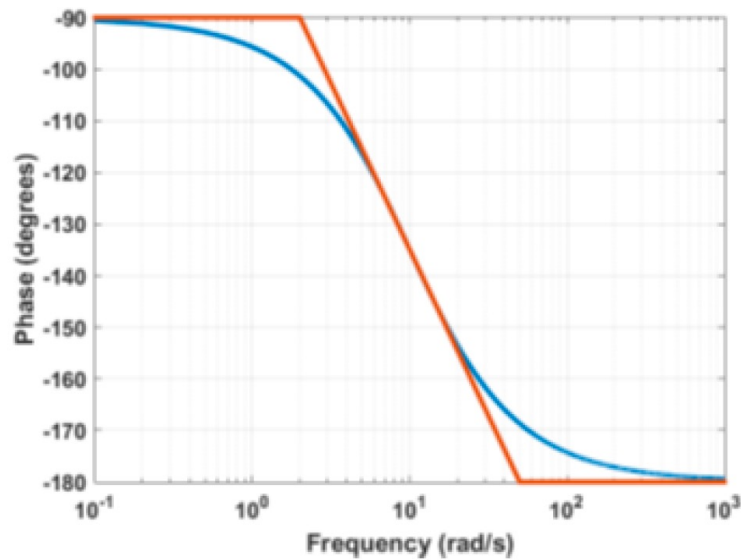
(b) The polar locus does not cross the negative real axis. Therefore, the gain margin is infinite (we can add as much gain as we want, and it will not change the stability of the closed loop system).

(c) To find the asymptotic Bode diagrams, we should first identify the simple terms in H. We have:

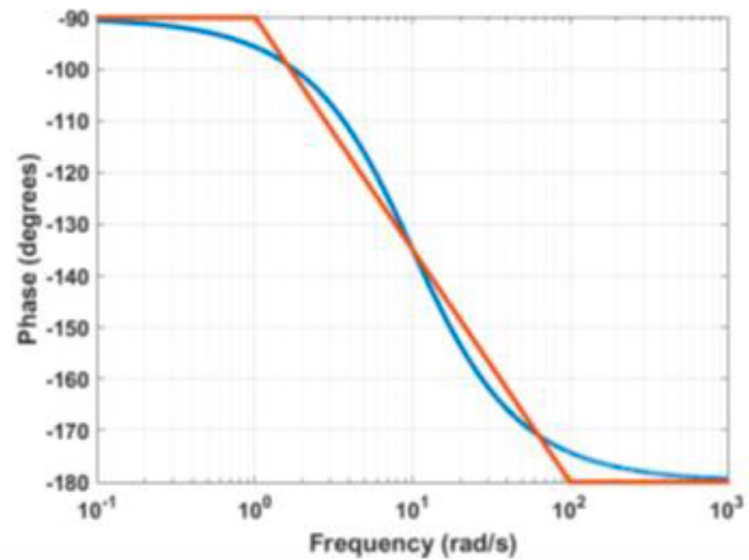
- a gain of 5 or 14dB
- a pole at origin or integrator: $\frac{1}{s}$
- a simple lag of corner frequency 10: $\frac{1}{1+\frac{s}{10}}$

Using the asymptotic plots of each term gives the asymptotic plots presented in figure 2b for the gain and 2c for the phase. The latter shows the asymptotic plot for both a factor of 5 (i.e. $0.2\omega_C$ and $5\omega_C$) and a factor of 10 (i.e. $0.1\omega_C$ and $10\omega_C$).





(a)



(b)

Figure 2c: phase Bode diagram for H – real (blue) and asymptotic (orange). (a) using factor of 5 for asymptotic plot and (b) using a factor of 10.

Worked example 3

3. Nyquist stability:

- a) State the Nyquist stability criterion. Explain what is meant by *gain* and *phase margins*.
- b) Sketch a Nyquist plot of the open-loop frequency response of a closed-loop stable system, and indicate the gain and phase margins.
- c) Sketch a Bode plot of the open-loop frequency response of a closed-loop stable system, and indicate the gain and phase margins.

Solutions:

(a) The Nyquist stability criterion is:

A closed-loop system is stable if and only if the number of clockwise encirclements of the critical (-1) point by the polar locus, as ω goes from $-\infty$ to $+\infty$, is equal to the number of right half plane open loop zeros.

The Phase Margin Φ is equal to the extra phase lag that could be added to get closed loop instability. It is calculated by determining the phase, θ , when $|G(j\omega)| = 1$ and adding this phase to 180° .

$$\text{Phase Margin} = \Phi = 180^\circ + \theta$$

The Gain Margin, G_m , is defined as that factor by which you would have to increase the gain to get instability. It is calculated by determining the magnitude, λ , when $\arg G(j\omega) = 180^\circ$:

$$\text{Gain Margin} = \frac{1}{\lambda}$$

(b) Let:

$$G(s) = \frac{10}{(s+1)(s+2)(s+3)}$$

The Nyquist plot is shown on fig Q3b.

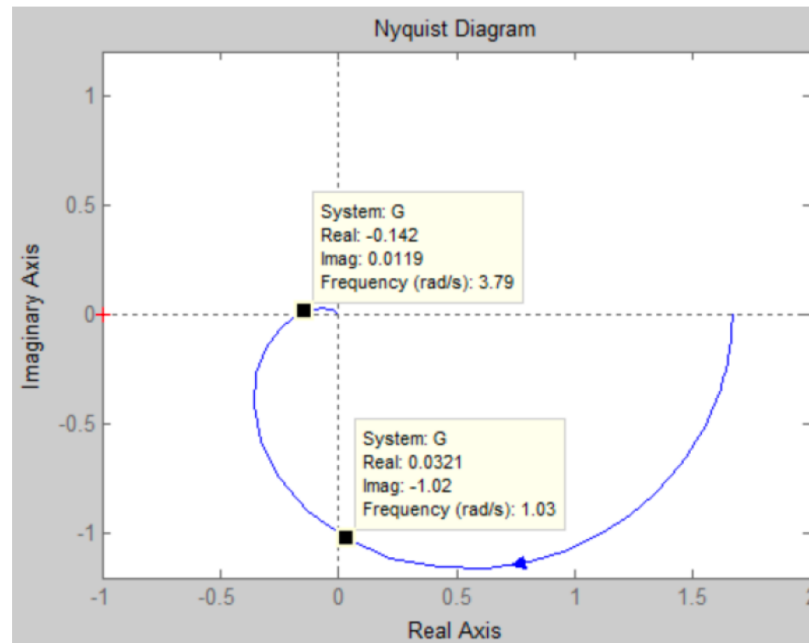


Figure Q3b: Example of Nyquist plot for a stable closed-loop system.

As can be seen from the plot, $\lambda = 0.15$

$$\Rightarrow \text{Gain Margin} = \frac{1}{\lambda} = 6.667$$

The phase at the frequency, such that the gain is equal to 1, is $\theta = -90^\circ$. Therefore, we have

$$\text{Phase Margin} = \Phi = 180^\circ + \theta = 90^\circ$$

(c) Let consider the same transfer function as in (b). Its Bode diagrams are below:

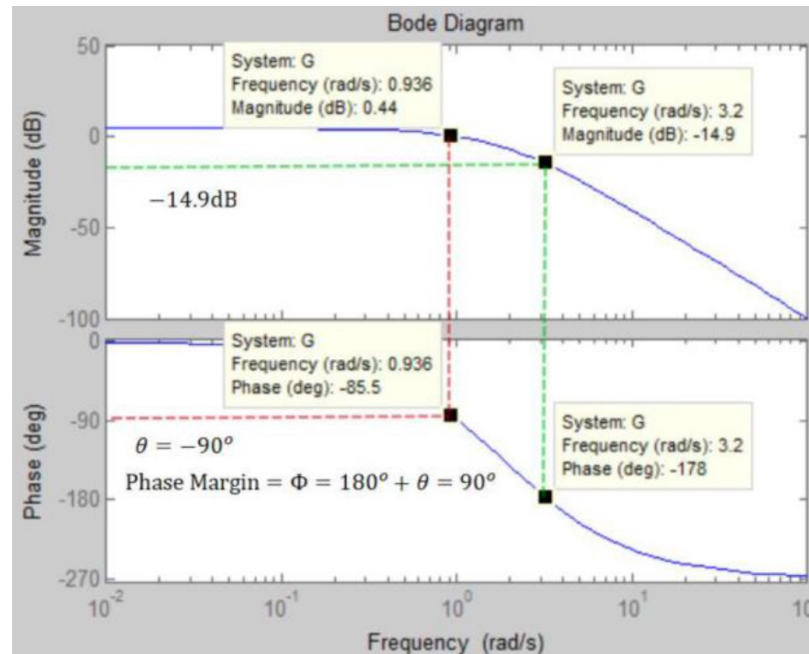


Figure Q3c: Example of Bode diagrams for a stable closed-loop system.

The gain margin can be calculated in decibel. We have:

$$\text{Gain Margin} = 14.9\text{dB}$$

We can convert this value in linear scale and we get:

$$\text{Gain Margin} = 10^{\frac{14.9}{20}} = 5.56$$

The difference in the value is due to the imprecision of the value read in the Matlab plot. As can be seen on fig. Q3c, the gain is obtained for a frequency where the phase is close to -180° but not exactly -180° .

Worked example 4

(4) Consider the following open loop transfer function:

$$G(s) = \frac{1}{s(s+2)^2}$$

Answer the following questions:

- (a) Express the transfer function in the frequency domain
- (b) Find the expression for the real and imaginary parts
- (c) What is the behaviour of G when $\omega \rightarrow 0$
- (d) Find the pulsation/frequency for which the polar locus crosses the negative real axis (i.e. $\text{Im}(G(j\omega)) = 0$)
- (e) Calculate the Gain margin

Solutions:

(a) Let's replace s by $j\omega$ in the expression of the open loop transfer function:

$$G(j\omega) = \frac{1}{j\omega(j\omega + 2)^2} = \frac{1}{-4\omega^2 + j\omega(4 - \omega^2)}$$

(b) To get the real and imaginary parts from the previous expression, we have to multiply both numerator and denominator by the complex conjugate of the denominator:

$$G(j\omega) = \frac{1}{-4\omega^2 + j\omega(4 - \omega^2)} = \frac{(-4\omega^2 - j\omega(4 - \omega^2))}{(-4\omega^2 + j\omega(4 - \omega^2))(-4\omega^2 - j\omega(4 - \omega^2))}$$
$$G(j\omega) = \frac{-4\omega^2 - j\omega(4 - \omega^2)}{16\omega^4 + \omega^2(4 - \omega^2)^2}$$

We can deduce:

$$\begin{aligned}\operatorname{Real}[G(j\omega)] &= -\frac{4\omega^2}{16\omega^4 + \omega^2(4 - \omega^2)^2} = -\frac{4}{16\omega^2 + (4 - \omega^2)^2} \\ \operatorname{Imag}[G(j\omega)] &= -\frac{\omega(4 - \omega^2)}{16\omega^4 + \omega^2(4 - \omega^2)^2} = \frac{(4 - \omega^2)}{\omega(16\omega^2 + (4 - \omega^2)^2)}\end{aligned}$$

(c) When $\omega \rightarrow 0$, we have:

$$\begin{cases} \operatorname{Re}[G(j\omega)] \rightarrow -\frac{1}{4} \\ \operatorname{Im}[G(j\omega)] \rightarrow -\infty \end{cases}$$

This is a vertical asymptote at $\operatorname{Real} = -\frac{1}{4}$

(d) We start with $\operatorname{Im}(G(j\omega)) = 0$. Therefore,

$$\frac{(4 - \omega^2)}{\omega(16\omega^2 + (4 - \omega^2)^2)} = 0 \Rightarrow 4 - \omega^2 = 0 \Rightarrow \omega = \pm 2$$

As we are working with positive frequencies, the solution is $\omega = 2$.

Now, we can calculate the value of the real part (i.e. the position of the locus crossing the negative real axis) by injecting the value of frequency we found above into the real part (we could also use the gain). We have:

$$\operatorname{Re}[G(j2)] = -\frac{1}{16}$$

λ is the absolute value of the real part, therefore $\lambda = \frac{1}{16}$.

(e) The gain margin is the inverse of λ : *Gain Margin* = $1/\lambda = 16$

Worked example 5

For the following frequency response functions of the open-loop systems sketch a Nyquist diagram and use it to predict whether or not the corresponding closed-loop system would be stable

a) $G(j\omega) = \frac{10}{(1 + j\omega)^2}$

b) $G(j\omega) = \frac{8}{j\omega(1 + j\omega/5)(1 + j\omega/10)}$

c) $G(j\omega) = \frac{10}{(j\omega)^2(1 + j\omega/10)}$

d) $G(j\omega) = \frac{10(1 + j\omega/2)}{(j\omega)^2(1 + j\omega/10)}$

e) $G(j\omega) = \frac{4(1 + j\omega)^2}{(j\omega)^2(1 + 5j\omega)}$

What would happen here if the gain were reduced by a factor of 3?

[Hint: consider the behaviour at low and high frequencies and where $\arg G(j\omega) = -180^\circ$ ($\text{Imag}\{G(j\omega)\} = 0$).]

A general method (that may sometimes need to be adapted to specific cases) is:

- 1- Find the frequency response of the open loop by substituting s by $j\omega$;
- 2- Express the real/imaginary parts and the gain/phase;
- 3- Find the point for $\omega=0$ or $\omega \rightarrow 0$, if 0 is a pole of the transfer function;
- 4- Find the limit of the Nyquist locus when $\omega \rightarrow +\infty$;
- 5- Find the point on the Nyquist locus for each corner frequencies in the transfer function.

(a) The open loop transfer function is a double lag term. Therefore, the corner frequency is $\omega = 1$. We already have the frequency response, so let's find the real and imaginary parts.

$$G(j\omega) = \frac{10}{(1 + j\omega)^2} = \frac{10}{1 - \omega^2 + j2\omega} = \frac{10(1 - \omega^2 - j2\omega)}{(1 - \omega^2)^2 + 4\omega^2}$$

$$\Rightarrow \begin{cases} \text{Re}[G(j\omega)] = \frac{10(1 - \omega^2)}{(1 - \omega^2)^2 + 4\omega^2} \\ \text{Im}[G(j\omega)] = -\frac{20\omega}{(1 - \omega^2)^2 + 4\omega^2} \end{cases}$$

The gain is:

$$|G(j\omega)| = \frac{10}{\sqrt{(1 - \omega^2)^2 + 4\omega^2}}$$

The phase is not as straightforward. The imaginary part is always negative; therefore, the Nyquist plot is in the bottom half of the plane. However, the real part can be positive or negative depending on the value of ω :

- If $\omega < 1$: the real part is positive and the imaginary part is negative. The Nyquist locus is the 4th quadrant and the phase is:

$$\angle G(j\omega) = -\tan^{-1} \frac{2\omega}{1 - \omega^2}$$

- If $\omega > 1$: the real part is now negative; the imaginary part is still negative. Therefore, the locus is in the third quadrant and the phase expression is:

$$\angle G(j\omega) = \tan^{-1} \frac{2\omega}{1 - \omega^2} - 180^\circ$$

We can see that when $\omega = 1$, the phase is 90° .

Let's now consider the three key frequencies we have identified:

- $\omega = 0$: we have

$$\begin{cases} |G(j0)| = 10 \\ \angle G(j0) = 0^\circ \end{cases}$$

- $\omega \rightarrow +\infty$: we have

$$\begin{cases} |G(j\omega)| \rightarrow 0 \\ \angle G(j\omega) \rightarrow -180^\circ \end{cases}$$

- $\omega = 1$: we have

$$\begin{cases} |G(j1)| = 5 \\ \angle G(j1) = -90^\circ \end{cases}$$

The Nyquist plot is shown on fig 5a.

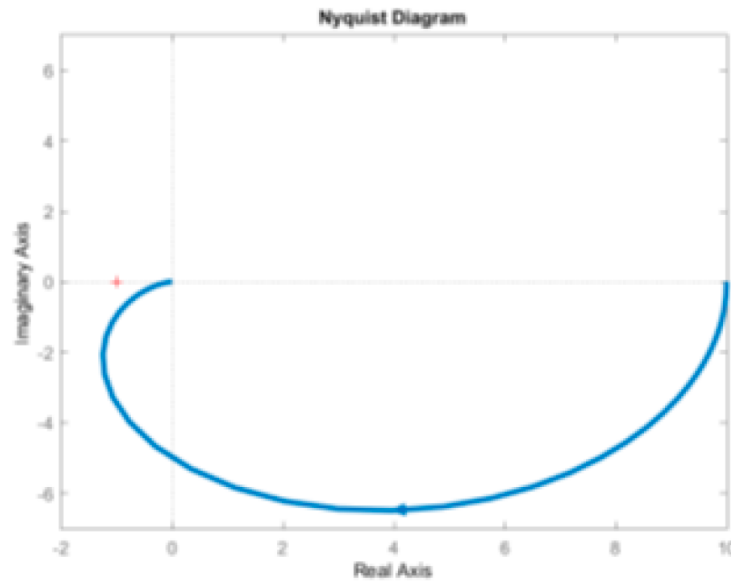


Figure 5a: Nyquist plot for question (a)

The Nyquist plot does not encircle the -1 point so the system is stable

Worked example 6

For the frequency response functions given in Question 4, sketch a Bode diagram (including magnitude and phase). Predict from the Bode diagram whether or not the corresponding closed loop system would be stable, and compare your answer with those obtained using the Nyquist diagrams.

(a) The transfer function is:

$$G(s) = \frac{10}{(1+s)^2}$$

It is made of a simple gain of 10 and two simple lag terms with corner frequency 1.

The gain in decibel is: 20dB. The asymptotic plot is easy to find as the simple lag terms give a slope of -40dB/decade from $\omega = 1$. For the phase, it varies from 0 to -180° and the value at $\omega = 1$ is 90° .

The Bode diagrams are presented in figure 6a.

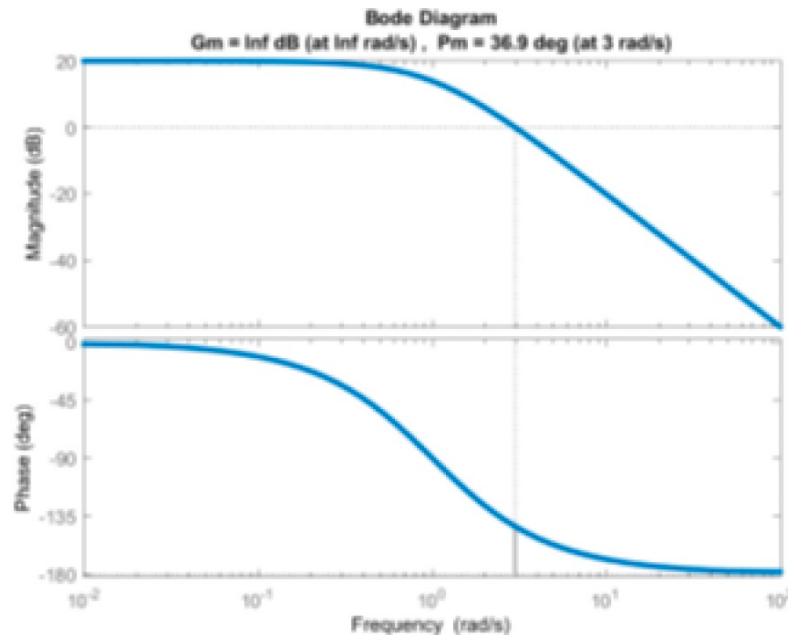


Figure 6a: Bode diagrams for question 6(a).

As the phase is never equal to -180° , the gain margin is infinite. For the phase margin, and as the phase is always above -180° , the phase margin is positive, and the closed-loop system is stable.

Worked example 7

An open-loop system has a transfer function

$$G(s) = \frac{K e^{-sT}}{s}$$

Draw a Nyquist plot for this system and show that the corresponding closed-loop system will be stable provided that $2KT < \pi$.

$$G(j\omega) = \frac{K e^{-j\omega T}}{j\omega} = \frac{K e^{-j(\omega T + \frac{\pi}{2})}}{\omega}$$

The gain is:

$$|G(j\omega)| = \frac{K}{\omega}$$

The phase can be read directly as the transfer function is a complex exponential. We have (be careful when manipulating the phase. The unit of the phase in the complex exponential is radians, so ωT is in radians):

$$\angle G(j\omega) = -\omega T - \frac{\pi}{2} = -\omega T \frac{180^\circ}{\pi} - 90^\circ$$

We can also express the real and imaginary parts. We get:

$$\begin{cases} \operatorname{Re}[G(j\omega)] = -\frac{K}{\omega} \sin \omega T \\ \operatorname{Im}[G(j\omega)] = -\frac{K}{\omega} \cos \omega T \end{cases}$$

To sketch the Nyquist plot, we should look for the behaviour when $\omega \rightarrow 0$ and $\omega \rightarrow \infty$.

- $\omega \rightarrow 0$: we have:

$$\begin{cases} |G(j\omega)| \rightarrow +\infty \\ \angle G(j\omega) = -90^\circ \\ \operatorname{Re}[G(j\omega)] = -KT \\ \operatorname{Im}[G(j\omega)] = -\infty \end{cases}$$

- $\omega \rightarrow +\infty$: this case is a bit more difficult, especially the phase. For the gain, we have:

$$|G(j\omega)| \rightarrow 0$$

The real and imaginary parts also tend toward 0.

For the phase, the value does not converge as it is proportional to ωT . Therefore, the phase is varying from 0 to 2π and repeats. Therefore, the Nyquist plot **spirals** towards 0 as $\omega \rightarrow +\infty$.

The limit of stability corresponds to when the system has a phase of -180° and a gain of 1. Let's find the frequencies that gives a phase of -180° :

$$\begin{aligned} -180^\circ &= -90^\circ - \omega T \frac{180^\circ}{\pi} + m360^\circ \\ \Rightarrow \frac{\omega T}{\pi} &= \frac{1}{2} + 2m \\ \Rightarrow \omega &= \frac{\pi}{2T} + m \frac{2\pi}{T} \end{aligned}$$

Which frequency to select? The gain is proportional to $1/\omega$. Therefore, if the stability condition is guaranteed for the lowest frequency (i.e. the highest gain), it will also be guaranteed for the other frequencies. Therefore, we have:

$$\omega_s = \frac{\pi}{2T}$$

Let's inject this frequency into the expression of the gain of the open loop:

$$|G(j\omega_s)| = \frac{K}{\omega_s} = \frac{2KT}{\pi}$$

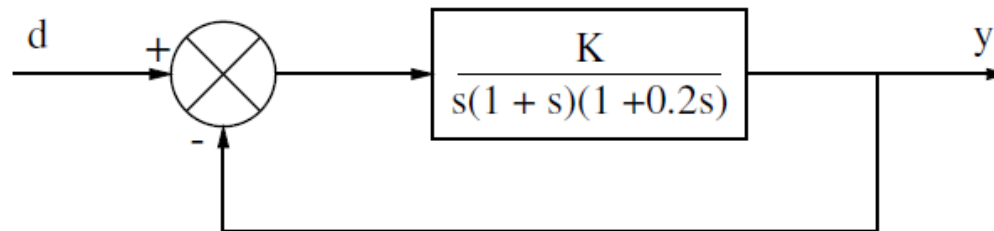
To have the magnitude at this frequency less than 1, the condition on K and T must be:

$$\frac{2KT}{\pi} < 1 \Rightarrow 2KT < \pi$$

Worked example 7

What value of gain K will give the system shown below

- i. a phase margin of 60° ?
- ii. a gain margin of 12 dB?



Solutions:

(a) The open-loop transfer function in the frequency response is:

$$G(s) = \frac{5K}{j\omega(j\omega + 1)(j\omega + 5)} = \frac{5K}{j\omega(5 - \omega^2 + j6\omega)} = \frac{5K}{-6\omega^6 + j\omega(5 - \omega^2)}$$

We can deduce the real and imaginary parts of G by multiplying numerator and denominator by the complex conjugate of the denominator:

$$\begin{cases} \operatorname{Re}[G(j\omega)] = -\frac{30K\omega^2}{36\omega^4 + \omega^2(5 - \omega^2)^2} \\ \operatorname{Im}[G(j\omega)] = -\frac{5K\omega(5 - \omega^2)}{36\omega^4 + \omega^2(5 - \omega^2)^2} \end{cases}$$

From the frequency response and, the real and imaginary parts, we can find the gain and phase of G :

$$\begin{cases} |G(j\omega)| = \frac{5K}{\sqrt{36\omega^4 + \omega^2(5 - \omega^2)^2}} \\ \angle G(j\omega) = \tan^{-1}\left(\frac{5 - \omega^2}{6\omega}\right) - 180^\circ \end{cases}$$

(a) In order to have a phase margin of 60° , the phase of G , θ , should be:

$$\text{Phase Margin} = 180^\circ + \theta = 60^\circ \Rightarrow \theta = -120^\circ$$

Let's use the general expression of the phase of G to find the frequency that gives a phase of -120° .

$$\theta = -120^\circ = \tan^{-1}\left(\frac{5 - \omega^2}{6\omega}\right) - 180^\circ$$

$$\begin{aligned}
 \theta &= -120^\circ = \tan^{-1} \left(\frac{5 - \omega^2}{6\omega} \right) - 180^\circ \\
 &\Rightarrow \tan^{-1} \left(\frac{5 - \omega^2}{6\omega} \right) = 60^\circ \\
 &\Rightarrow \frac{5 - \omega^2}{6\omega} = \tan 60^\circ = \sqrt{3} \\
 &\Rightarrow \omega^2 + 6\sqrt{3}\omega - 5 = 0 \\
 &\Rightarrow \begin{cases} \omega_1 = 0.461 \\ \omega_2 = -10.853 \end{cases}
 \end{aligned}$$

As we are only considering positive frequencies, the solution is:

$$\omega_1 = 0.461$$

Now that the frequency is known, in order to find the value of K giving the phase margin of 60° , we use the expression of the gain. From the definition of the phase margin, we know that the gain must be 1 at the frequency giving the phase margin. Therefore, we must have:

$$\begin{aligned}
 |G(j\omega_1)| &= 1 \Rightarrow \frac{5K}{\sqrt{36\omega_1^4 + \omega_1^2(5 - \omega_1^2)^2}} = 1 \\
 &\Rightarrow 25K^2 = 36\omega_1^4 + \omega_1^2(5 - \omega_1^2)^2 \\
 &\Rightarrow K^2 = \frac{\omega_1^6 + 26\omega_1^4 + 25\omega_1^2}{25} = 0.2595
 \end{aligned}$$

Therefore, the value of K is:

$$K = 0.5094$$

(b) In this question, we are looking for the value of K that gives a gain margin of 12dB, or 3.98 in linear gain. The gain margin is defined as the inverse of the gain when the phase is -180° . Therefore, the first step is to find the frequency that gives a phase of -180° .

We have:

$$\begin{aligned}\angle G(j\omega) &= \tan^{-1} \left(\frac{5 - \omega^2}{6\omega} \right) - 180^\circ = -180^\circ \\ \Rightarrow \tan^{-1} \left(\frac{5 - \omega^2}{6\omega} \right) &= 0 \\ \Rightarrow \frac{5 - \omega^2}{6\omega} &= 0 \Rightarrow 5 - \omega^2 = 0 \\ \Rightarrow \omega_{1,2} &= \pm\sqrt{5}\end{aligned}$$

As we are only considering positive frequencies, the solution is:

$$\omega_1 = \sqrt{5}$$

We can now calculate the gain, λ , at this frequency. We have:

$$\lambda = |G(j\omega_1)| = \frac{5K}{\sqrt{36\omega_1^4 + \omega_1^2(5 - \omega_1^2)^2}} = \frac{K}{6}$$

Using the definition of the gain margin, we get:

$$\text{Gain Margin} = \frac{1}{\lambda} = \frac{6}{K}$$

We want this gain margin to be 3.98, therefore K must be:

$$\frac{6}{K} = 3.98 \Rightarrow K = 1.507$$